Detects gases such as ethylene from 1-10ppm.

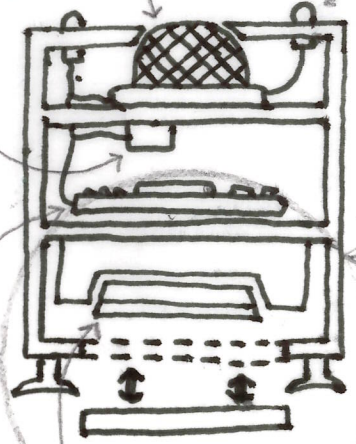
Fruit ripening releases ethylene and it can reach up to 1-10ppm when spoiling

MQ-135 particle sensor

Buzzer

Red LEDs

LED's flash and buzzer turns on when elevated amounts of particles such as ethylene are detected by over-ripened fruit. This gives you a chance to remove it before it accelerates the spoiling of other fruit



Food-safe outer casing be 3D-printed

Customizable

Could → Easy prototyping

Biodegradable

Cheap

OR, oven heated acrylic could be shaped around a wooden former

More suited for a final product

Rounded edges, safer

Tough

Food-safe

Micro processor

Cell battery

OR

Might need to be 3Dprinted

Difficult to make hinge on small scale

Screw-in system

Better seal

Maybe not viable for a bridge

Also reduced wastage on corners

Aesthetically pleasing round design

No hinge needed

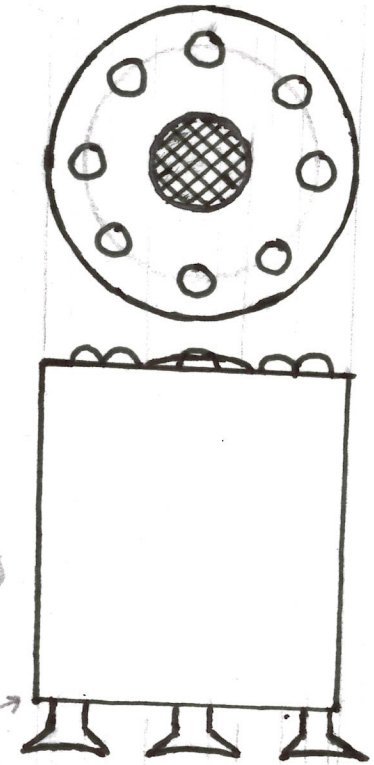
Flap system

No 'lock' needed, friction holds it in place

Can be laser cut

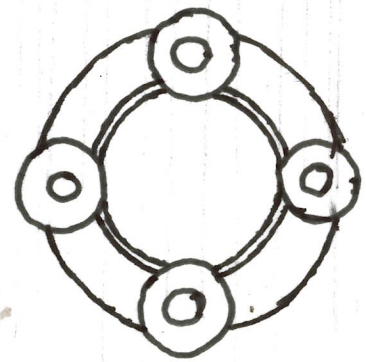
Would need a 'lock' to secure it in place

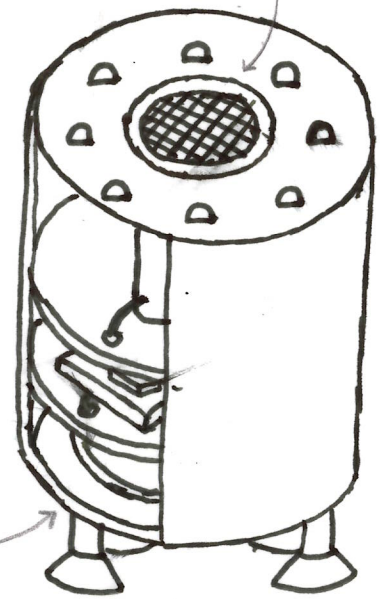
Could be difficult to grip or raised edges could snap off



Can be attached to virtually any smooth surface inside a fridge

Insulating the gap between the particle sensor and casing will be difficult





Idea 2

Expensive